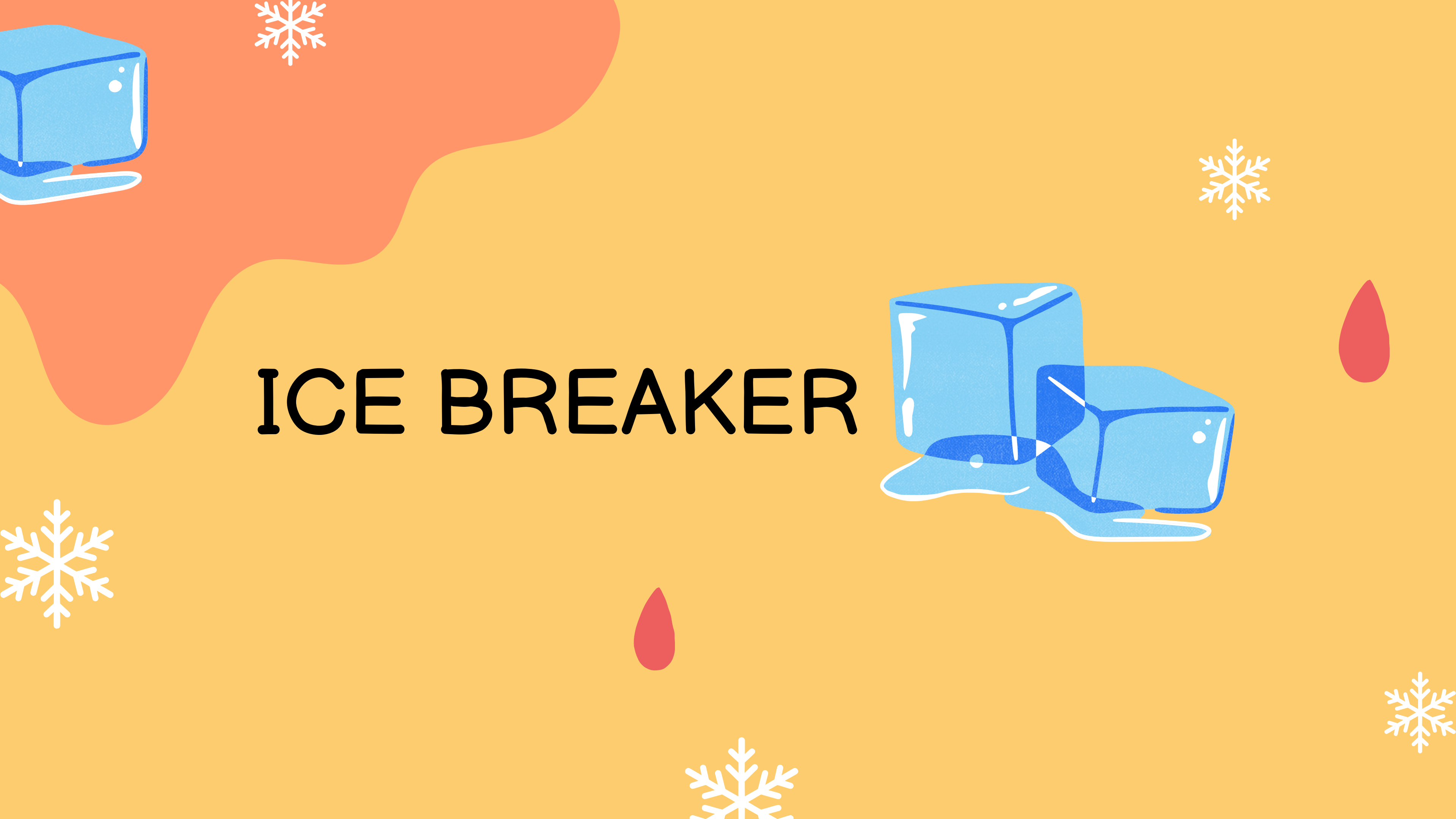
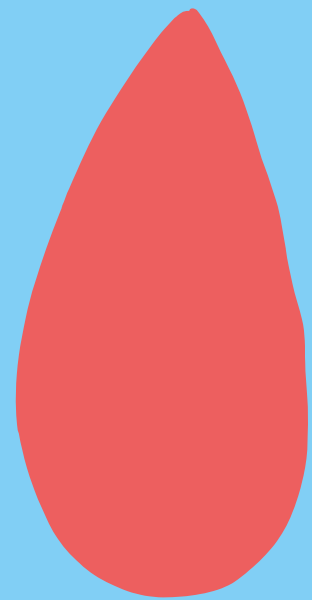




ICE BREAKER



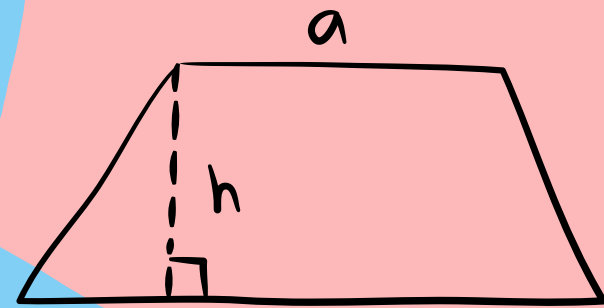
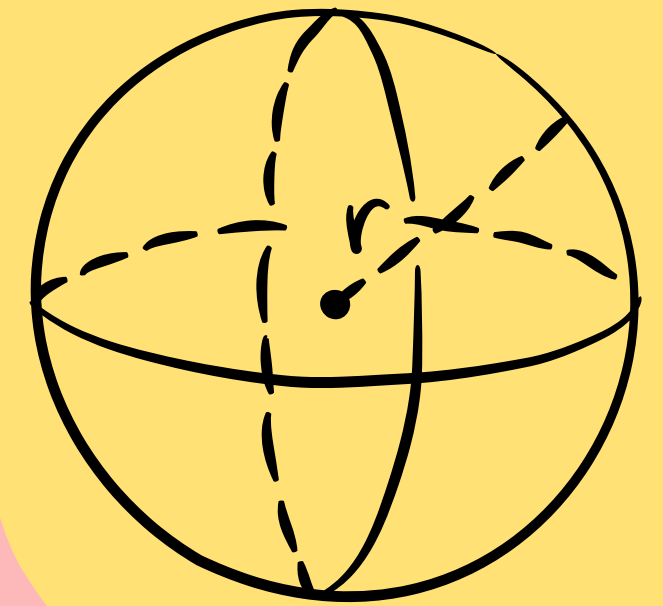
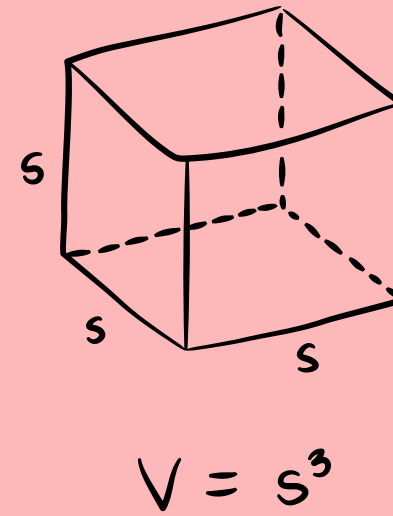
LET US DIVIDE TO FOUR
GROUPS



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

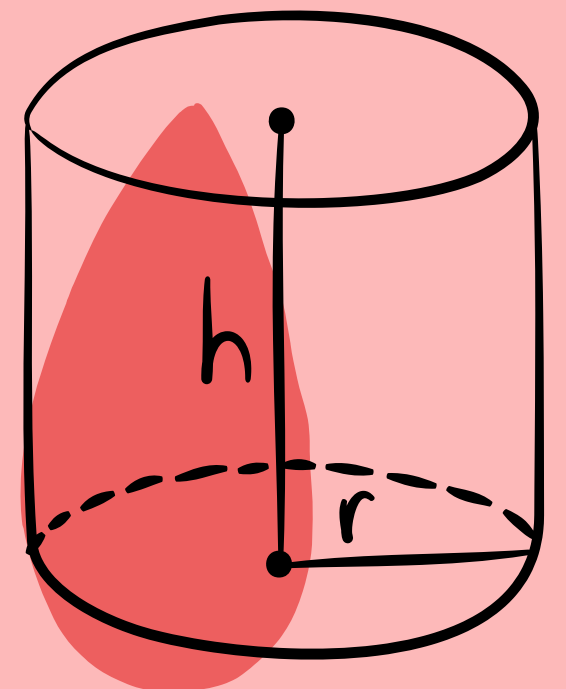
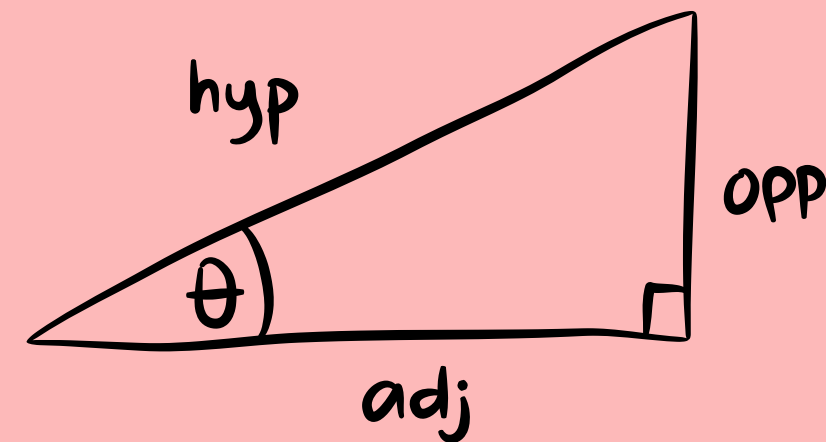
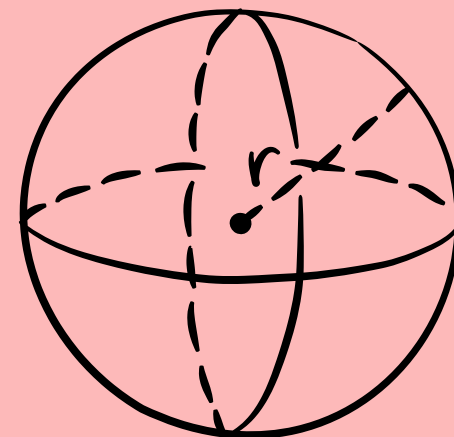
$$ax + by = c$$

$$ax^2 + bx + c = 0$$



Write me down:
Engineering formula
edition

$$\frac{x}{a} + \frac{y}{b} = 1$$



$$ax^2 + bx + c = 0$$

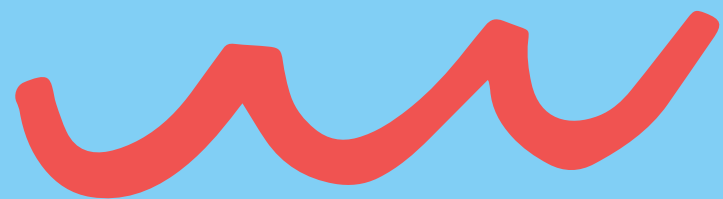
Area of a circle

$$\pi r^2 \quad \text{or} \quad \frac{\pi d^2}{4}$$



Circumference

$$2\pi r \text{ or } \pi d$$



Volume of a cylinder

$$\pi r^2 h$$



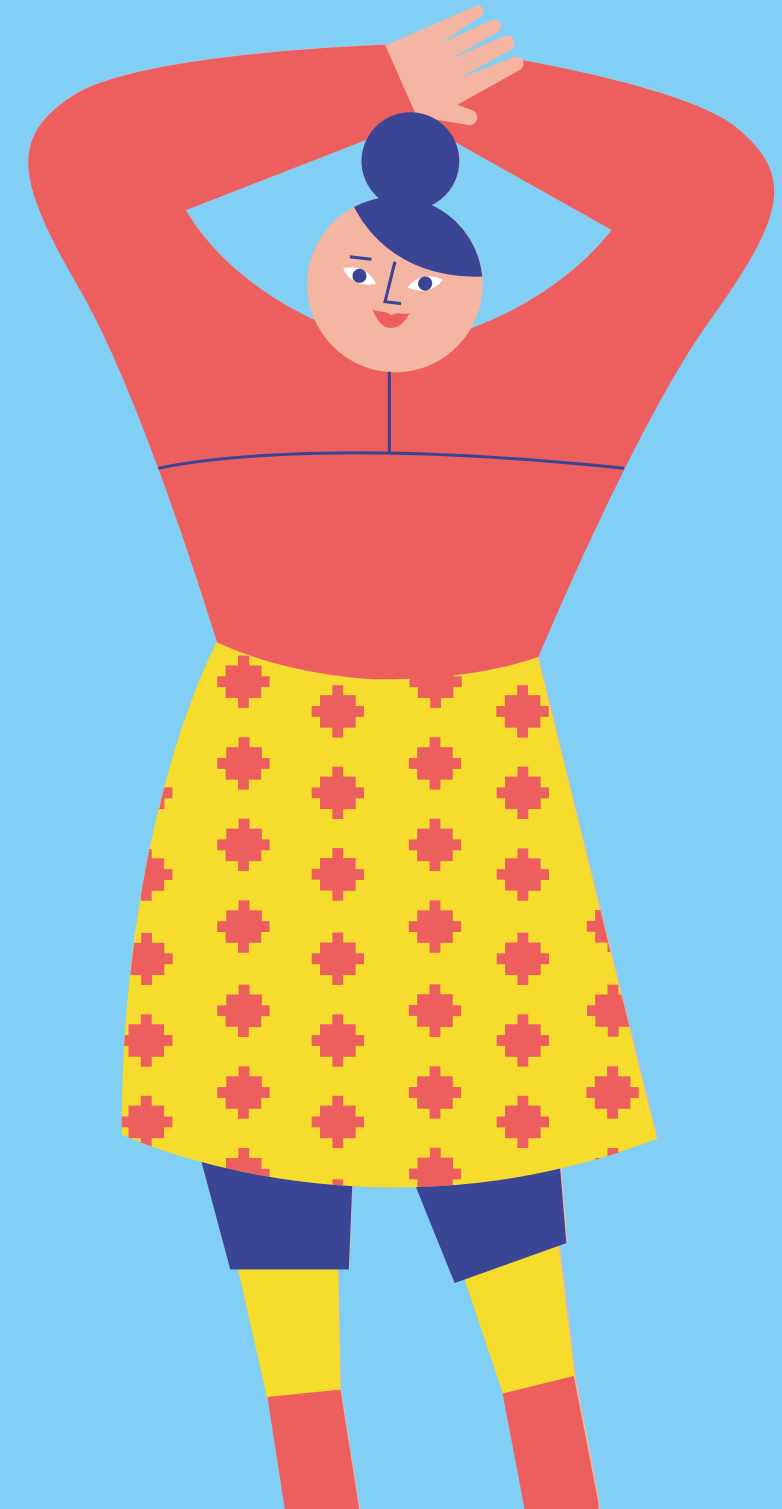
Area of a triangle

$$\frac{1}{2}bh$$



Newton's Second Law

$$Q = hA \Delta T$$



Pythagorean Theorem



$$c^2 = a^2 + b^2$$

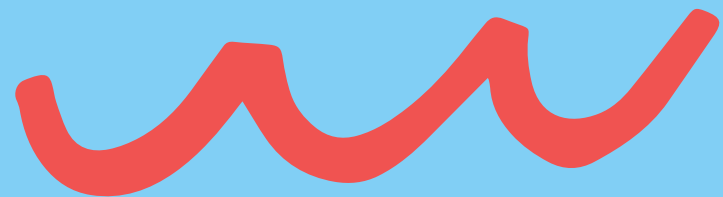
Kinetic Enegery

$$K = \frac{1}{2} m v^2$$



Ohm's Law

$$V = IR$$



Sensible heat transfer

$$Q = m C_p \Delta T$$



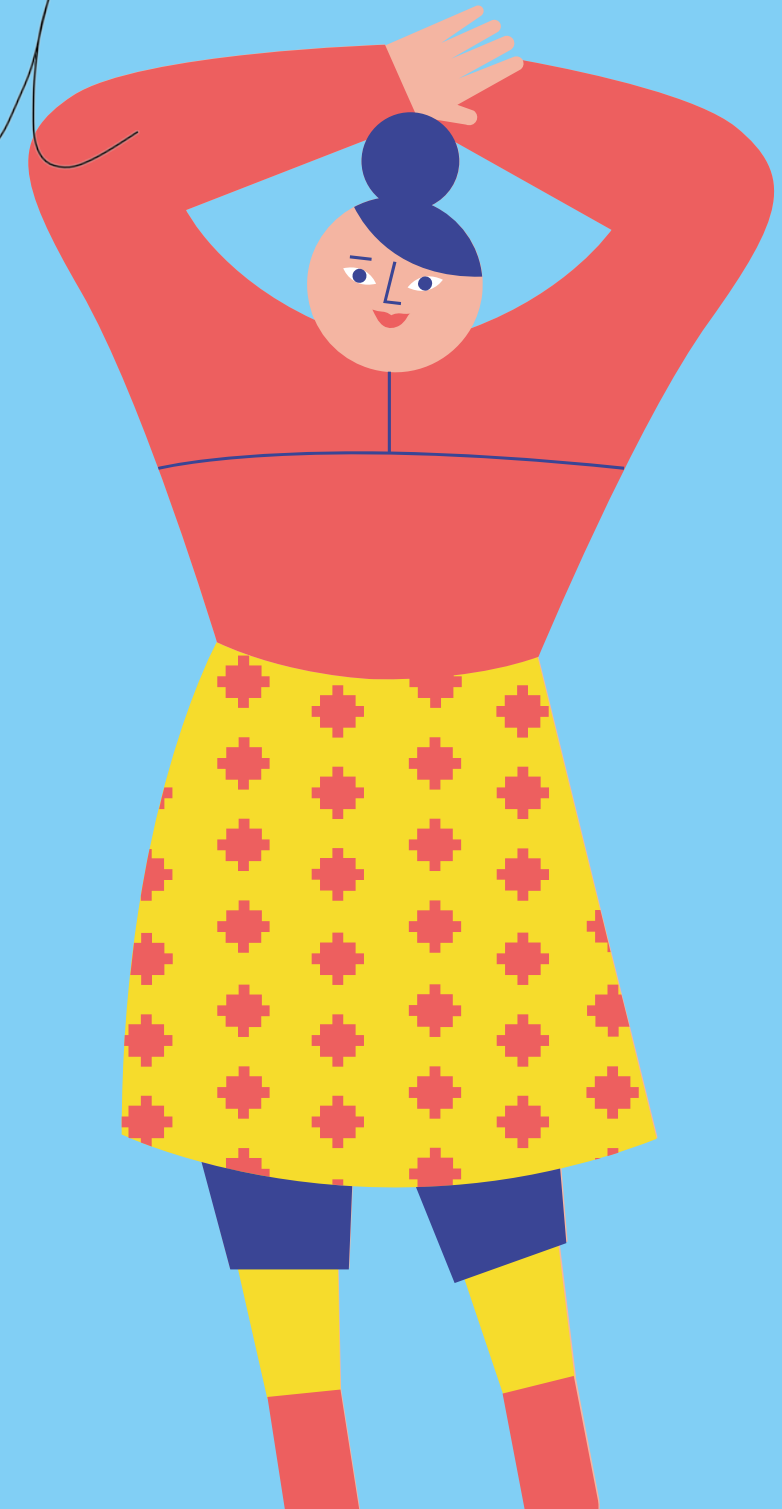
Momentum

$$p = mv$$



Reynold's number

$$Re = \frac{\rho V L}{\mu}$$



Hydraulic power

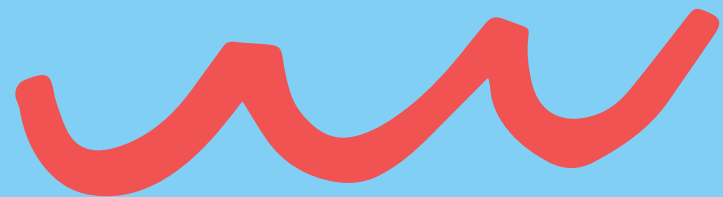


$$P = \rho g H Q$$



Electrical power

$$P = IV$$



Potential Energy

$$PE = mgh$$



Velocity

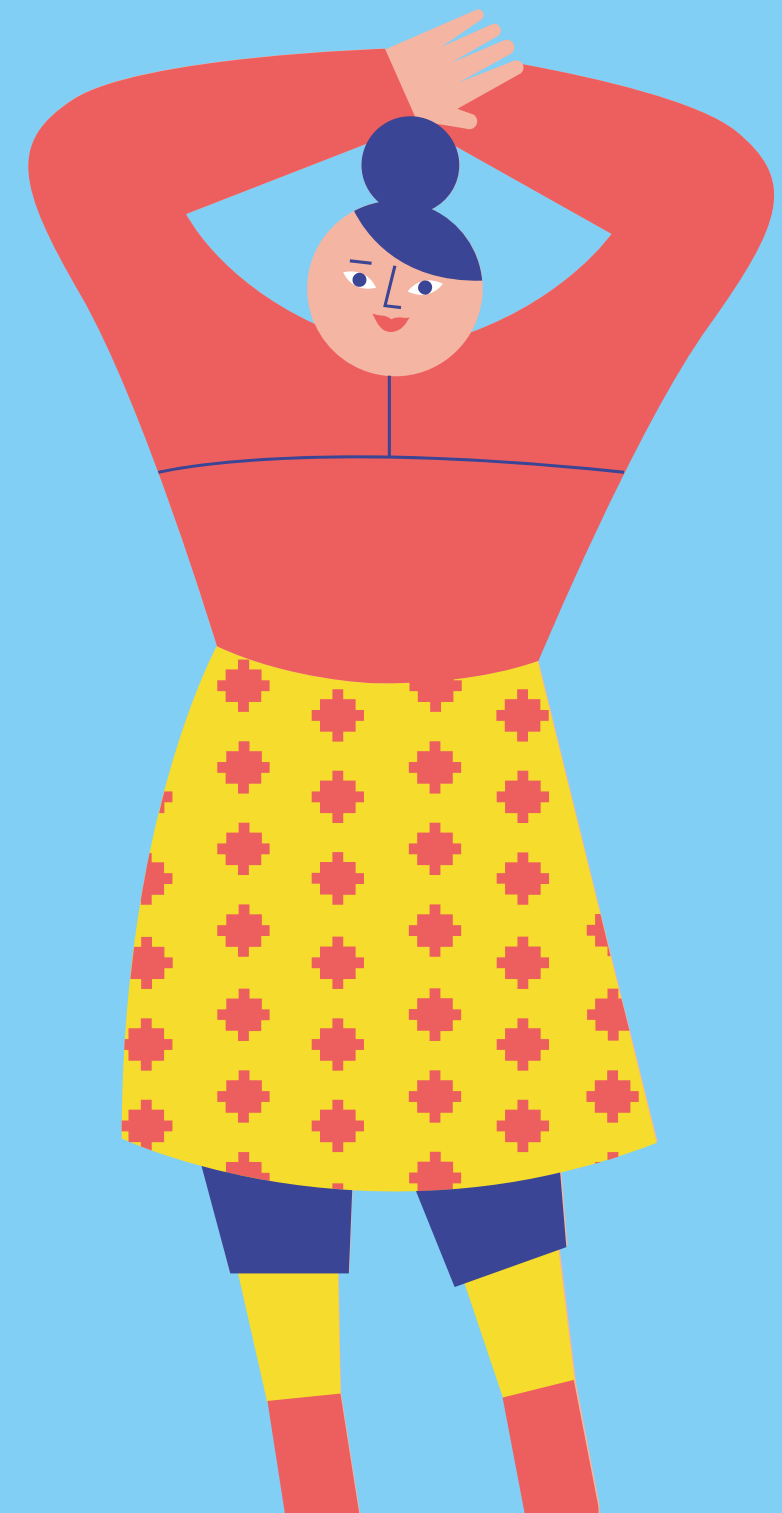
$$v =$$

$$\frac{d}{t}$$



Acceleration

$$a = \frac{d}{t^2}$$



Distance formula



$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



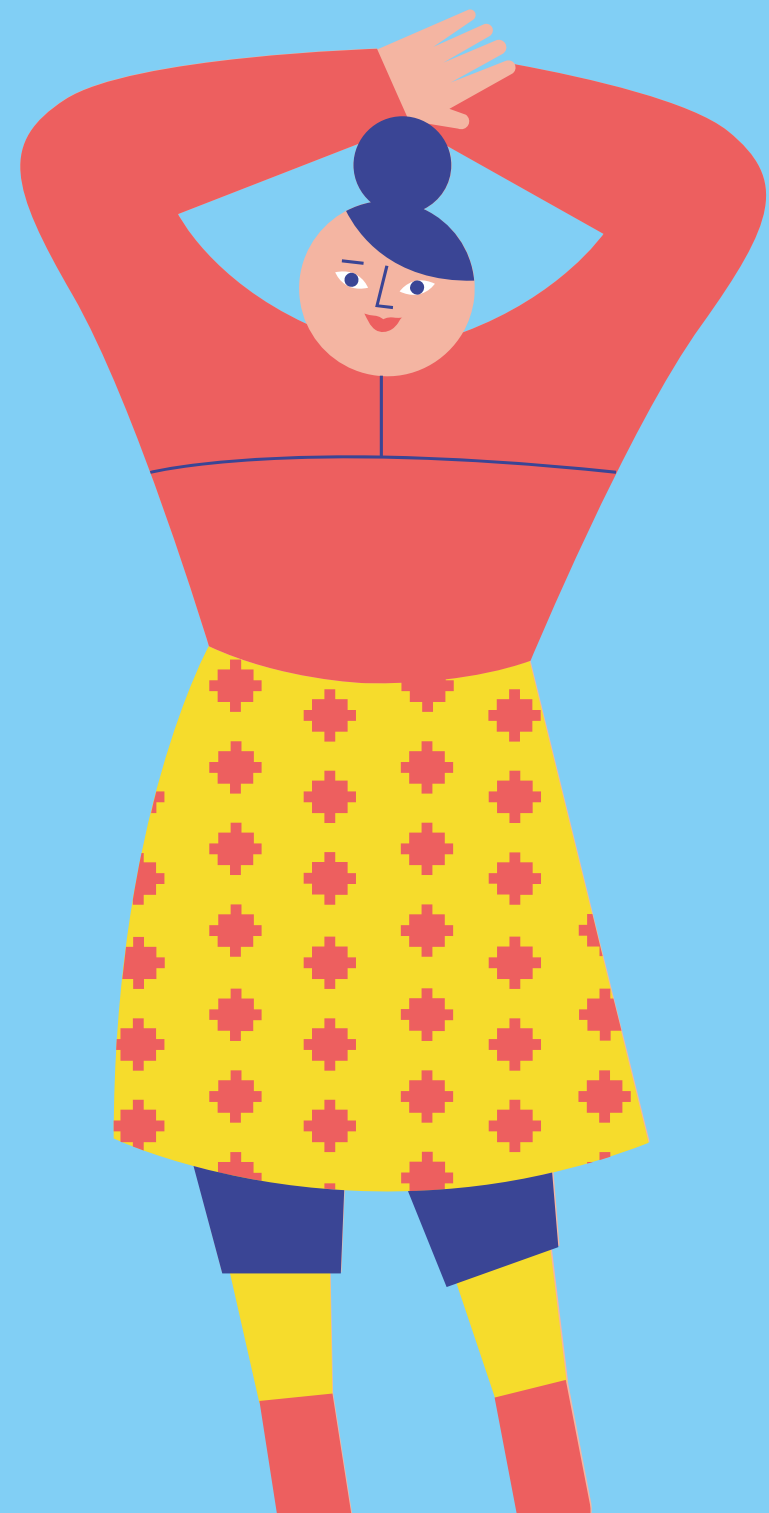
Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



Volume of a sphere

$$V = \frac{4}{3} \pi r^3$$



Combined Gas Law



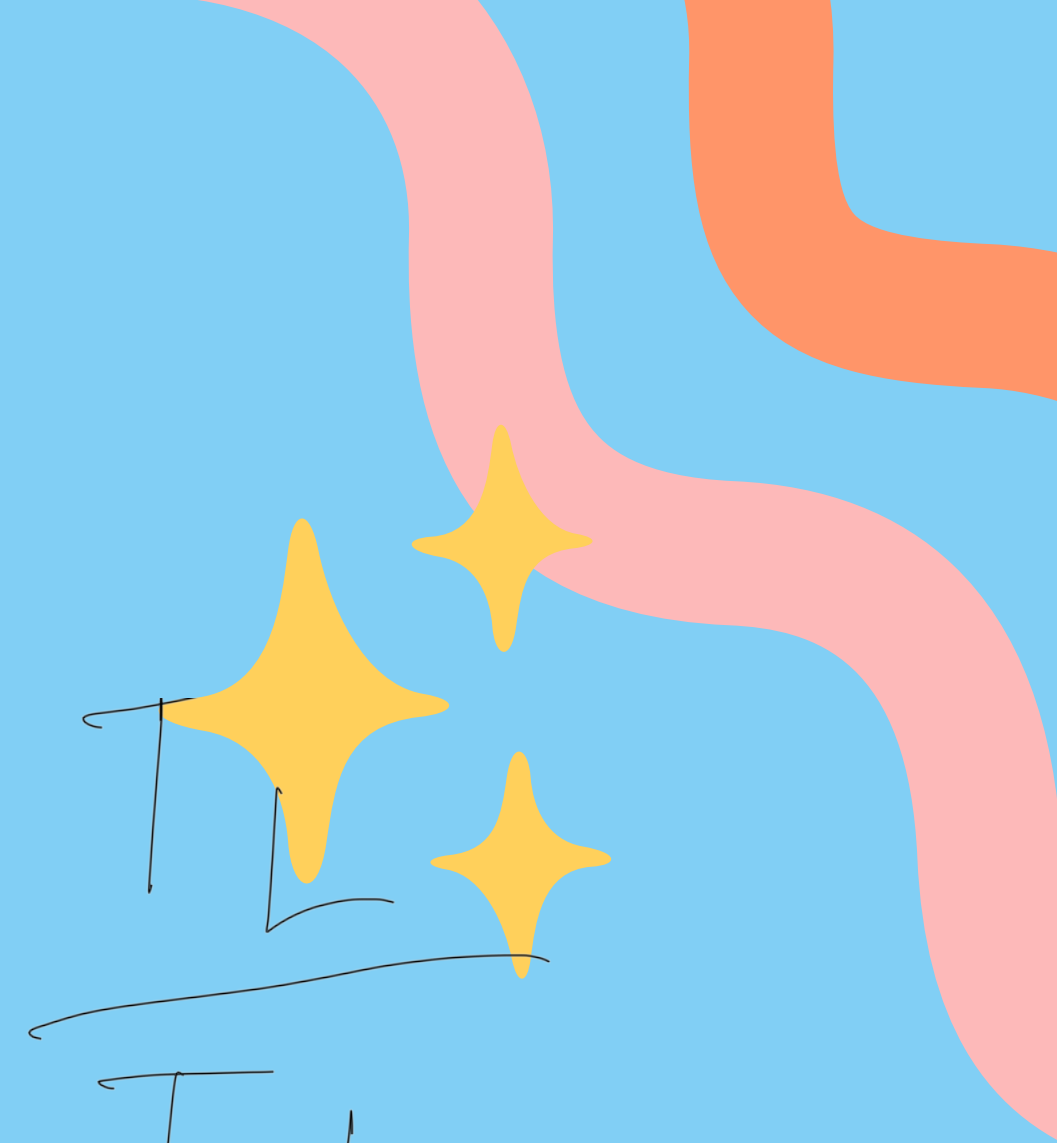
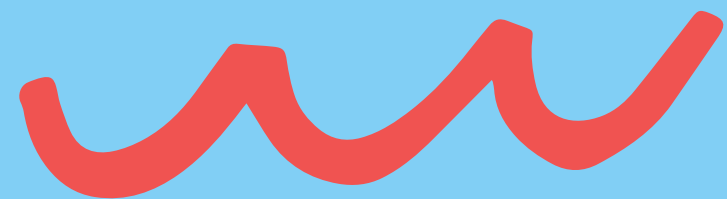
$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

Carnot

efficiency

$$\eta = 1 -$$

$$\frac{T_C}{T_H}$$



Capacitance

$$F = \frac{Q}{V}$$

capacitive
reactance

$$X_C = \frac{1}{2\pi f C}$$



Capacitance

$$F = \frac{Q}{V}$$

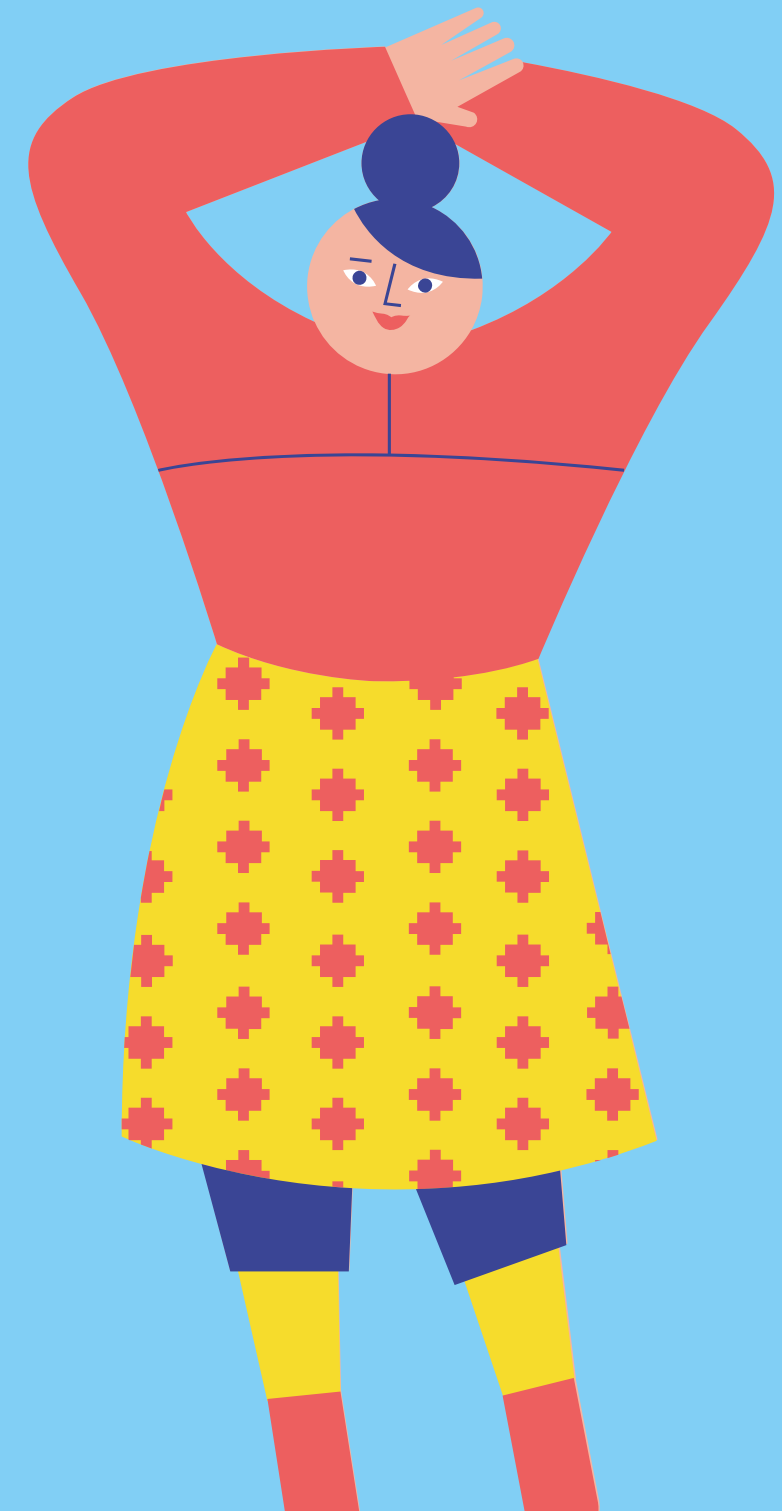
Stress

$$\sigma = \frac{F}{A}$$



Latent heat

$$Q = mL$$



Boyle's Law



$$P_1 V_1 = P_2 V_2$$



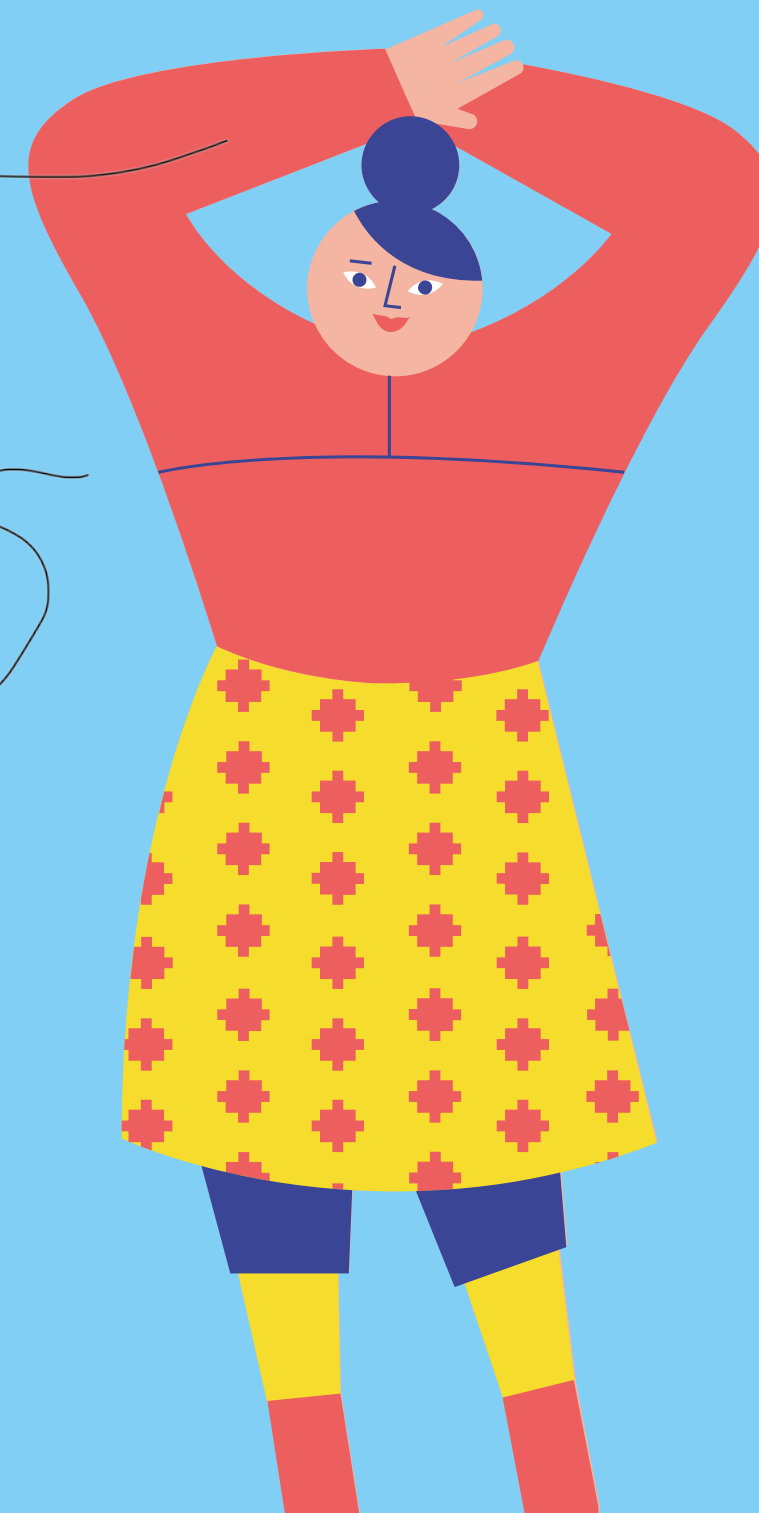
Convective Heat transfer

$$Q = h A \Delta T$$



Relative humidity

$$RH = \frac{P}{P_s}$$



Humidity ratio



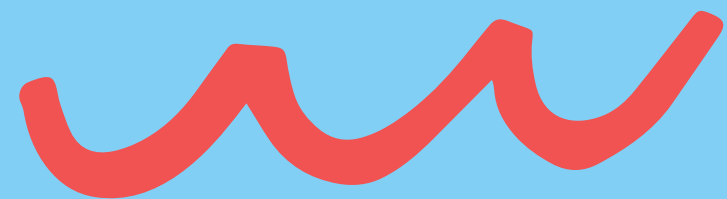
$$HR = \frac{m_v}{m_a}$$



First law of thermodynamics



$$\Delta U = Q - W$$



Enthalpy

$$h = u + pV$$



Radiation (Heat transfer)

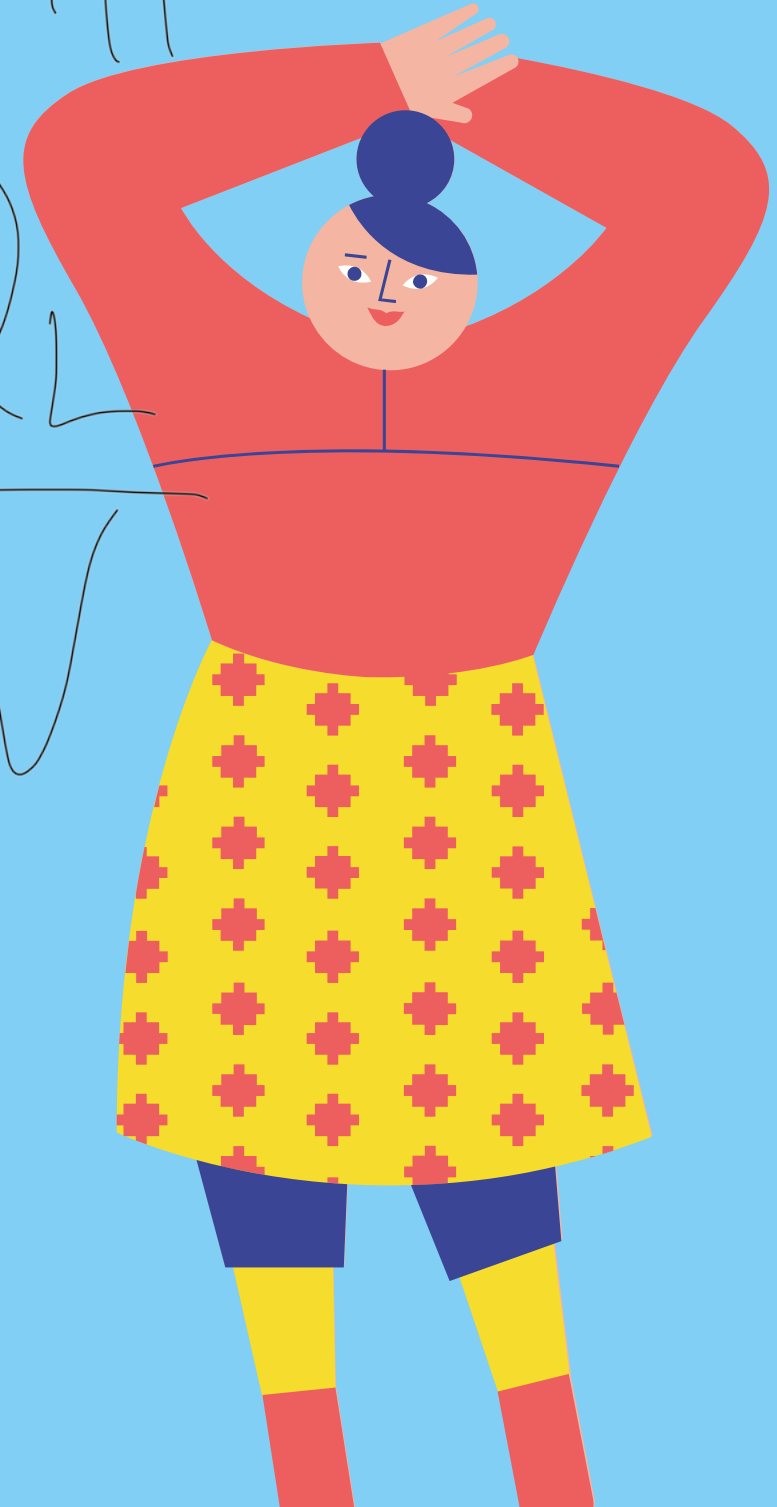
$$Q = \epsilon \sigma A \Delta T^4$$



COP of Ref

$$\text{COP} = \frac{Q_L}{Q_H - Q_L} = \frac{Q_L}{W}$$

$$\text{COP} = \frac{T_L}{T_H - T_L}$$



COP of heat pump



$$\text{COP} = \frac{Q_H}{Q_H - Q_L}$$



Conductive heat transfer

$$Q = \frac{KA\Delta T}{x}$$





Volume of sphere

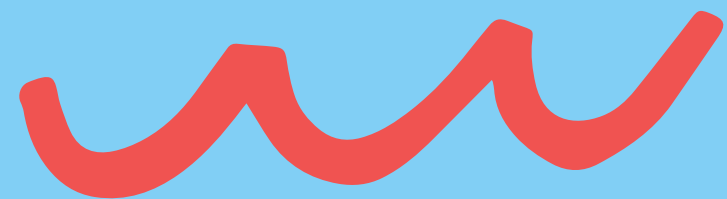
Final weight (moisture)



$$FW = IW \left[\frac{1 - MC_i}{1 - MC_F} \right]$$

MC wb to MC(db)

$$MC_{db} = \frac{MC_{wb}}{1 - MC_{wb}}$$



MC db to MC wb

$$MC_{wb} = \frac{MC_{db}}{1 + MC_{db}}$$



Degree C to degree F

$$^{\circ}\text{F} = ^{\circ}\text{C} \left(\frac{9}{5} \right) + 32$$



Gay Lussac's Law

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$



Charles's Law



$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$



Moisture Ratio

$$MR = \frac{MC_f - EMC}{MC_i - EMC}$$





Rectangle perimeter

$$P = 2W + 2l$$

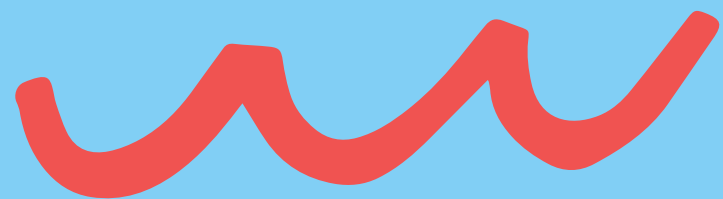
Density



$$\rho = \frac{m}{V}$$



Velocity



Acceleration



Frequency

$$F = \frac{1}{t}$$



Pipe Outer surface area

$$A = 2\pi r L$$



Pump efficiency



$$\eta_{\text{eff}} = \left(\frac{D_{\text{out}}}{D_{\text{in}}} \right) \times 100$$



Thank you for participating in
our icebreaker session,

Engineers!!!

